

DATA SCIENCE CAPSTONE PROJECT REPORT

SpaceX Falcon 9 First-Stage Landing Prediction

Prepared by: Kunal Singh
Submission date: 13 September 2026
GitHub: <https://github.com/kunalji-ji/data-science-capstone-kunal-singh>

Screenshot Evidence and Submission Guide

This document identifies the actual evidence screenshots that should be inserted into the final capstone report. Screenshots must be captured from the completed IBM notebooks and the working dashboard. Do not use sample images as evidence.

[-] PARTIAL

criterion_0

[PARTIAL] **Question 1: Did the learner uploaded the URL of GitHub repository, including all the completed notebooks and Python files Note:Note: Evaluate the submitted PDF to ensure it includes completed slides with relevant content for the following sections: Executive Summary, Introduction, Data Collection and Data Wrangling Methodology, EDA and Data Visualization Methodology, Predictive Analysis (Classification), EDA with Visualization, EDA with SQL Slides/Queries, Folium Map, Plotly Dash-related content, Predictive Analysis Results, and Conclusion. Slides must contain meaningful explanations, analysis, visuals, or results. Award 0 points if the slide contains only a heading, title, or keywords without supporting content. Question 2: Did the learner upload the final presentation as a .pdf file? Question 3: Did the learner complete the required Executive Summary slide? Question 4: Did the learner complete the required Introduction slide? Question 5: Did the learner complete the required data collection and data wrangling methodology slides? Question 6: Did the learner complete the required EDA and data visualization methodology slides? Question 7: Did the learner complete the predictive analysis (classification) slide? Question 8: Did the learner complete the required EDA with visualization slides? Question 9: Did the learner complete the required EDA with SQL slides/queries? Question 10: Did the learner complete the required Folium map slides? Question 11: Did the learner complete the required Plotly Dash-related slides? Question 12: Did the learner complete the predictive analysis results slides based on the following criteria? Question 13: Did the learner complete the Conclusion slide? The completed Conclusion slide should include at least: Question 14: Did the learner apply any creativity to this presentation? Creativity may include aspects such as: Question 15: Was the learner able to find and display any innovative insights? ** (2/15 points): The report provides a GitHub URL and identifies the intended repository, but it does not demonstrate that the repository contains completed notebooks and Python files. The PDF is largely a screenshot-evidence guide with explicit placeholder areas such as Insert the actual screenshot here, so the required completed project evidence is missing.

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1. Data Collection Screenshot

Capture the SpaceX API request, extracted response, and a real DataFrame preview such as df.head().

Suggested filename: 01_Data_Collection_API.png

Place under Data Collection Methodology.

EVIDENCE SCREENSHOT AREA
Insert the actual screenshot here after capturing it from your completed project.

2. Data Wrangling Screenshot

Capture cleaning code, missing-value handling, transformations, and df.info() or the cleaned DataFrame.

Suggested filename: 02_Data_Wrangling.png

Place under Data Wrangling Methodology.

EVIDENCE SCREENSHOT AREA
Insert the actual screenshot here after capturing it from your completed project.

3. EDA Visualization Screenshot

Capture an actual chart such as landing success by orbit, launch site, payload, or launch year. Include title and axis labels.

Suggested filename: 03_EDA_Visualizations.png

Place under EDA and Data Visualization.

EVIDENCE SCREENSHOT AREA
Insert the actual screenshot here after capturing it from your completed project.

4. SQL Query and Output Screenshot

Show an executed SQL query and its returned result table in the same screenshot. Query-only evidence is insufficient.

Suggested filename: 04_SQL_Queries_Results.png
Place under EDA with SQL.

EVIDENCE SCREENSHOT AREA

Insert the actual screenshot here after capturing it from your completed project.

5. Folium Map Screenshot

Show the actual interactive map with launch-site markers and, if possible, an open marker popup.

Suggested filename: 05_Folium_Launch_Sites_Map.png
Place under Folium Map.

EVIDENCE SCREENSHOT AREA

Insert the actual screenshot here after capturing it from your completed project.

6. Plotly Dash Screenshot

Show the running dashboard in a browser, including a dropdown or control and the chart/output it displays.

Suggested filename: 06_Plotly_Dash_Dashboard.png
Place under Plotly Dash.

EVIDENCE SCREENSHOT AREA

Insert the actual screenshot here after capturing it from your completed project.

7. Predictive Analysis Results Screenshot

Show the executed classification models and actual accuracy, precision, recall, F1-score, or model comparison output.

Suggested filename: 07_Predictive_Analysis_Results.png
Place under Predictive Analysis.

EVIDENCE SCREENSHOT AREA

Insert the actual screenshot here after capturing it from your completed project.

8. Confusion Matrix Screenshot

Show the actual confusion matrix with model name and predicted/actual class labels.

Suggested filename: 08_Confusion_Matrix.png
Place under Predictive Analysis Results.

EVIDENCE SCREENSHOT AREA

Insert the actual screenshot here after capturing it from your completed project.

9. GitHub Repository Screenshot

Show the repository name, URL, completed notebooks, app.py, README.md, and uploaded project files.

Suggested filename: 09_GitHub_Repository.png

Place in the appendix or repository section.

EVIDENCE SCREENSHOT AREA

Insert the actual screenshot here after capturing it from your completed project.

Final Submission Checklist

- All screenshots are from the learner’s own executed notebooks or working dashboard.
- Charts include meaningful titles, labels, and a short interpretation.
- SQL screenshots show both the query and output.
- Folium screenshot shows actual launch-site markers.
- Plotly Dash screenshot shows an interactive control and chart.
- Predictive results use real metrics copied from the notebook.
- GitHub repository is public and contains completed notebooks and Python files.
- No placeholder text such as 'insert screenshot here' remains in the submitted final PDF.

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"9. GitHub Repository Screenshot"

GRADING SUMMARY

Overall Correctness Score: **50%**

Feedback Breakdown:

[+] Correct: 0

[~] Partially Correct: 3

[X] Needs Improvement: 0

Detailed Feedback Summary:

How to Use This Feedback:

- * Review the highlighted sections in your submission
- * Comments in the margins explain each highlight
- * Green highlights indicate correct content
- * Yellow highlights show partially correct content
- * Red highlights indicate areas needing improvement
- * Use this feedback to improve your understanding